Quiz 5 Solutions

```r
library(factoextra)
```

```
## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.4.3

## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.4.1
```

```r
library(ggplot2)

# ------------------------------
# Data preparation
# ------------------------------
set.seed(123)
df <- read.csv("Student_Lifestyle_Data.csv")

# Remove missing values
clean_df <- na.omit(df)
nrow(clean_df)
```

```
## [1] 144
```

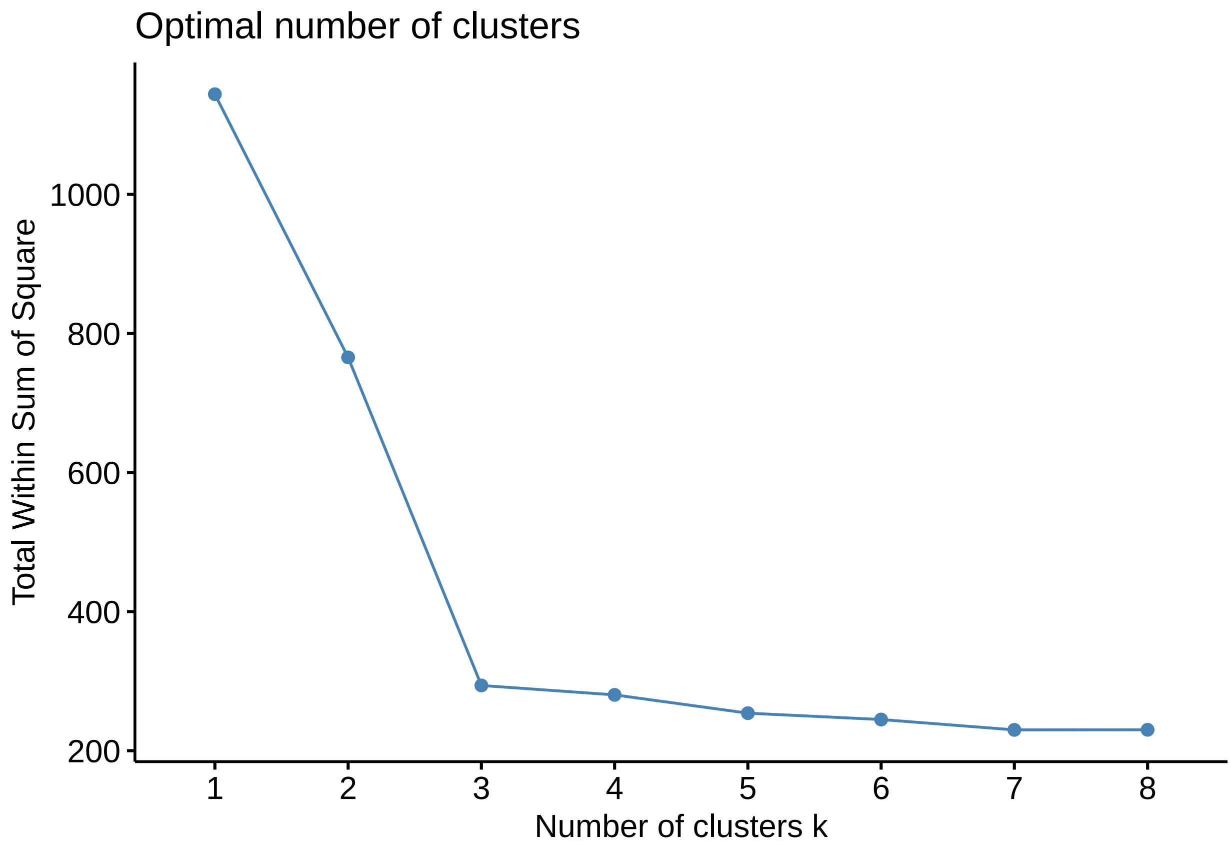
```r
# Exclude ID variable
x <- clean_df[, !(names(clean_df) %in% c("profile_id"))]

# Z-score standardization
x_scaled <- scale(x)

# Q1 comment:
# Scaling is important because clustering and PCA are sensitive to variable scale.
# Variables such as caffeine_mg and attendance_rate are on very different units,
# so unscaled data may let large-scale variables dominate the analysis.

# ------------------------------
# Task 1: Clustering
# ------------------------------

# Q2: WSS plot and K-means
fviz_nbclust(x_scaled, kmeans, method = "wss", k.max = 8)
```

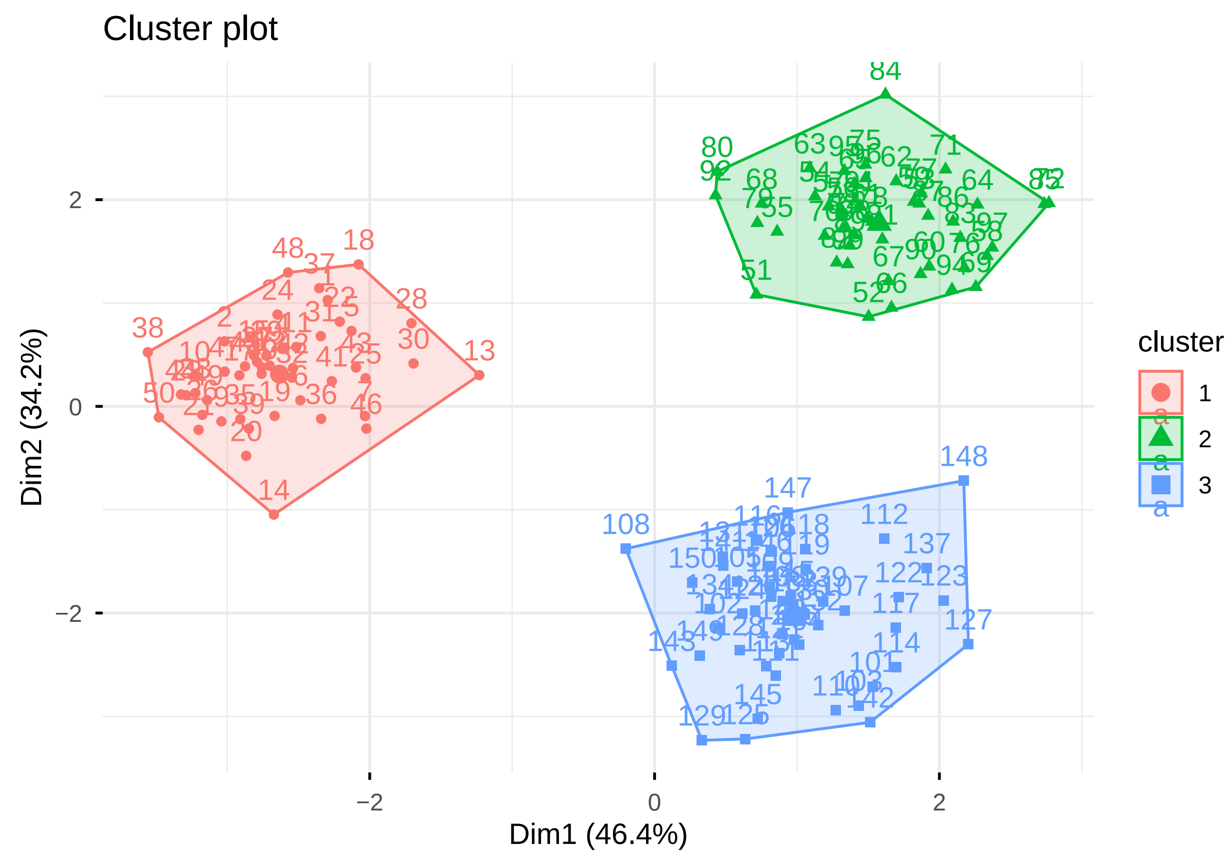



```r
# Choose k based on the elbow in the WSS plot.
# A reasonable choice here is k = 3.

set.seed(123)
km_scaled <- kmeans(x_scaled, centers = 3, nstart = 25)
km_scaled$size
```

```
## [1] 47 48 49
```

```r
# Q3: 2D cluster plot and cluster centers
fviz_cluster(km_scaled, data = x_scaled,
             ellipse.type = "convex",
             ggtheme = theme_minimal())
```




```r
round(km_scaled$centers, 3)
```

```
##   sleep_hours study_hours screen_time exercise_days stress_score caffeine_mg
## 1      -1.066       1.135       0.016        -0.445        1.086       1.163
## 2       0.950      -0.133      -1.079         1.259       -1.059      -0.824
## 3       0.092      -0.959       1.042        -0.806       -0.004      -0.308
##   social_hours attendance_rate
## 1       -1.130           0.774
## 2        0.048           0.389
## 3        1.036          -1.124
```

```r
# Brief interpretation should be based on the cluster centers.
# Cluster 1: lower sleep, higher study, higher stress, higher caffeine, lower social
# Cluster 2: higher sleep, more exercise, lower screen time, lower stress, lower caffeine
# Cluster 3: higher screen time, higher social time, lower attendance, lower study

# Q4: K-means on unscaled data
set.seed(123)
km_raw <- kmeans(x, centers = 3, nstart = 25)
km_raw$size
```

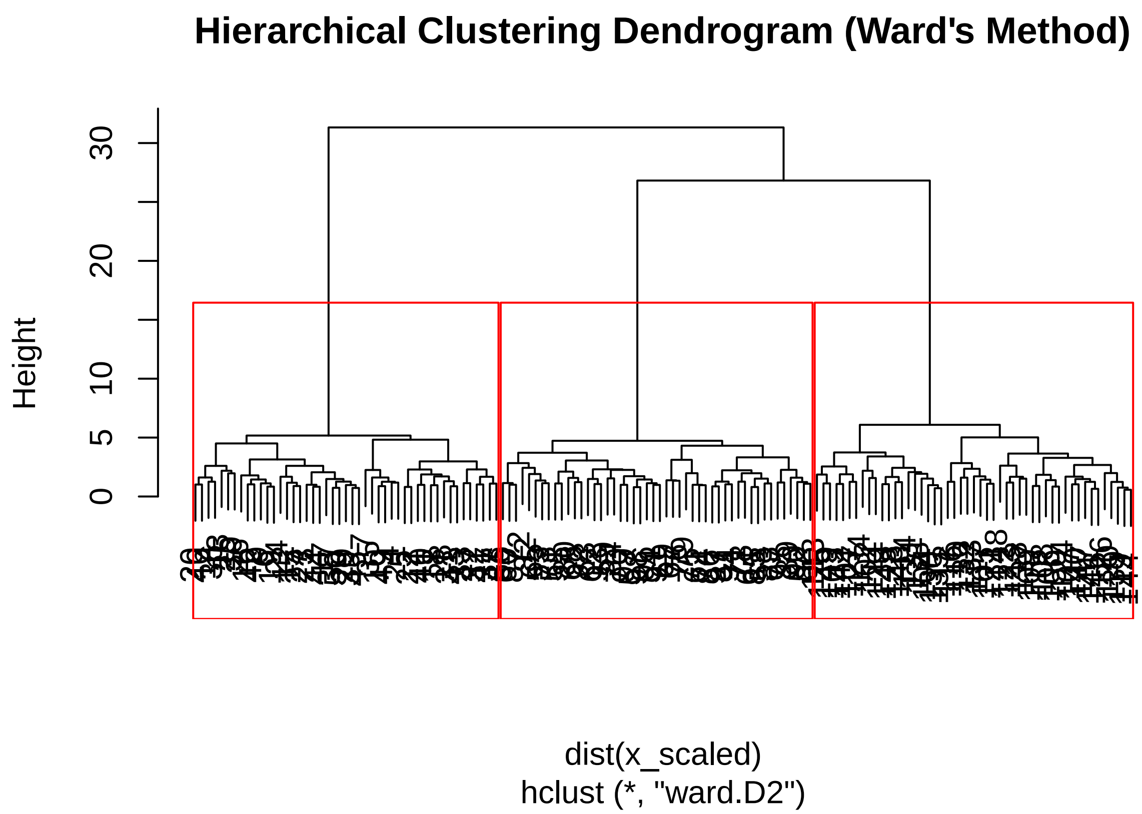
```
## [1] 40 65 39
```

```r
table(Scaled = km_scaled$cluster, Unscaled = km_raw$cluster)
```

```
##       Unscaled
## Scaled  1  2  3
##      1 39  7  1
##      2  0 18 30
##      3  1 40  8
```

```r
# Scaling often changes clustering because Euclidean distance is strongly influenced
# by variables with larger numerical ranges.
# The clustering assignments changed substantially after scaling, as shown by the contingency table

# Q5: Hierarchical clustering with Ward's method
hc_ward <- hclust(dist(x_scaled), method = "ward.D2")
plot(hc_ward, main = "Hierarchical Clustering Dendrogram (Ward's Method)")
rect.hclust(hc_ward, k = 3, border = "red")
```




```r
hc_groups <- cutree(hc_ward, k = 3)
table(hc_groups)
```

```
## hc_groups
##  1  2  3
## 47 48 49
```

```r
# Q6: Compare K-means and hierarchical clustering
table(Kmeans = km_scaled$cluster, Ward = hc_groups)
```

```
##        Ward
## Kmeans  1  2  3
##      1 47  0  0
##      2  0 48  0
##      3  0  0 49
```

```r
# -----------------------------
# Task 2: PCA
# -----------------------------

# Q7: PCA
pca_fit <- prcomp(x_scaled, center = TRUE, scale. = TRUE)
pca_summary <- summary(pca_fit)
pca_summary
```

```
## Importance of components:
##                          PC1    PC2     PC3     PC4    PC5     PC6     PC7
## Standard deviation     1.927 1.6539 0.63425 0.55624 0.5028 0.47716 0.44307
## Proportion of Variance 0.464 0.3419 0.05028 0.03868 0.0316 0.02846 0.02454
## Cumulative Proportion  0.464 0.8059 0.85621 0.89489 0.9265 0.95495 0.97949
##                            PC8
## Standard deviation     0.40509
## Proportion of Variance 0.02051
## Cumulative Proportion  1.00000
```

```r
# Proportion of variance explained by PC1-PC3
pve <- pca_summary$importance[2, 1:3]
cum_pve_3 <- pca_summary$importance[3, 3]
pve
```

```
##     PC1     PC2     PC3
## 0.46399 0.34194 0.05028
```

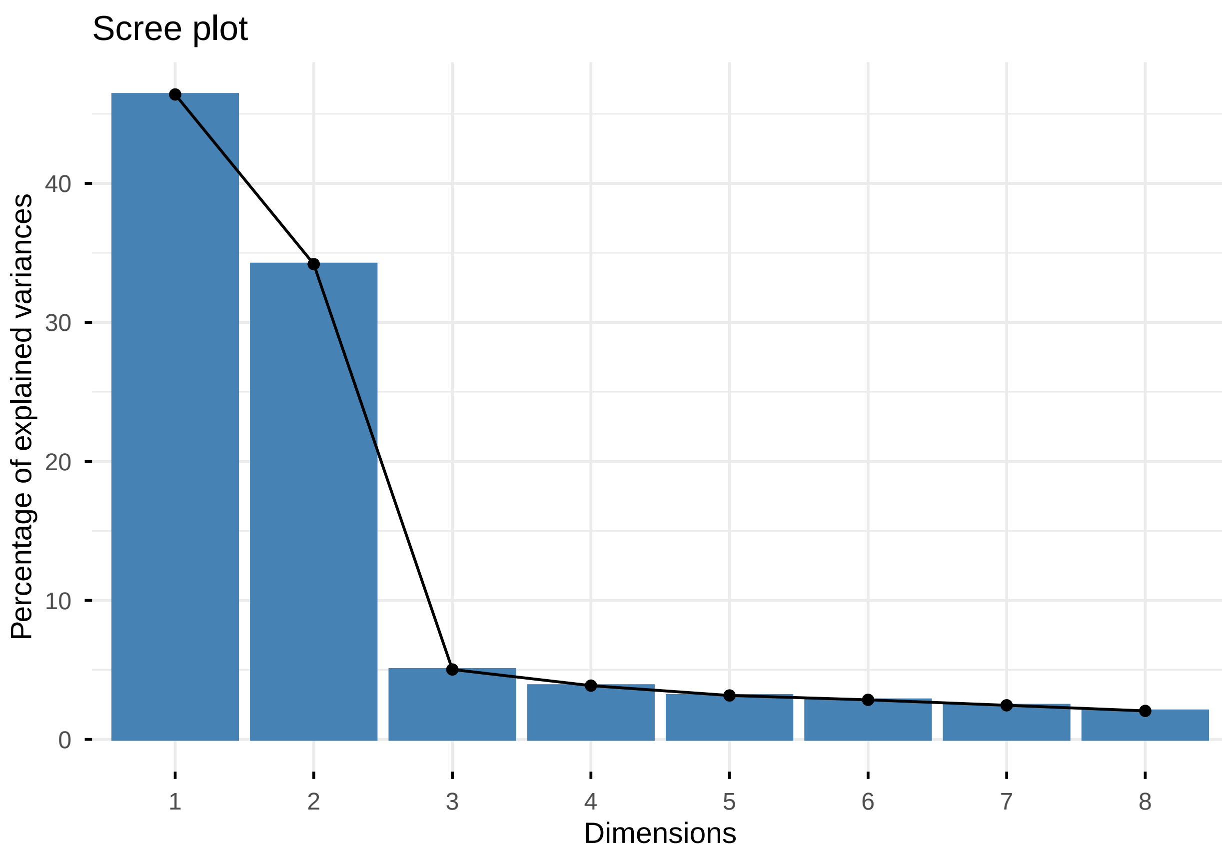
```r
cum_pve_3
```

```
## [1] 0.85621
```

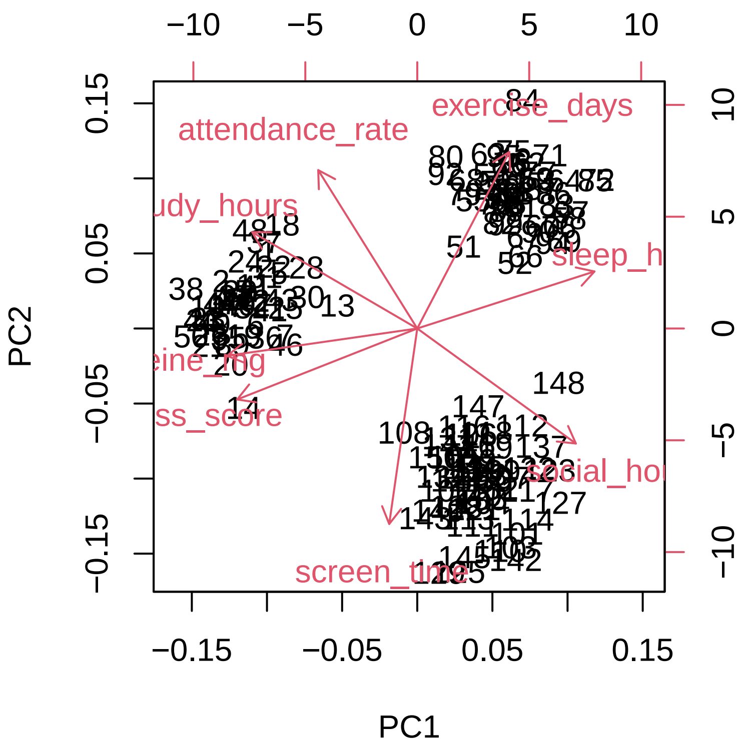
```r
# The first two principal components already explain about 80.6% of the total variation, so retaining t

# Q8: Scree plot and biplot
fviz_eig(pca_fit)
```

```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
```




```r
biplot(pca_fit)
```




```r
# Students should justify how many PCs to keep based on scree plot and cumulative PVE.
# The scree plot shows a clear drop after PC2, suggesting that PC1 and PC2 capture most of the dominant

# Q9: Largest absolute loadings and interpretation of PC1/PC2
round(pca_fit$rotation[, 1:2], 3)
```

```
##                    PC1    PC2
## sleep_hours      0.428  0.161
## study_hours     -0.398  0.271
## screen_time     -0.068 -0.551
## exercise_days    0.222  0.497
## stress_score    -0.435 -0.200
## caffeine_mg     -0.459 -0.077
## social_hours     0.383 -0.324
## attendance_rate -0.239  0.446
```

```r
pc1_abs <- sort(abs(pca_fit$rotation[, 1]), decreasing = TRUE)
pc2_abs <- sort(abs(pca_fit$rotation[, 2]), decreasing = TRUE)
pc1_abs
```

```
##     caffeine_mg    stress_score     sleep_hours     study_hours    social_hours
##       0.4593474       0.4348509       0.4279993       0.3984236       0.3830193
## attendance_rate   exercise_days     screen_time
##       0.2392210       0.2224096       0.0677009
```

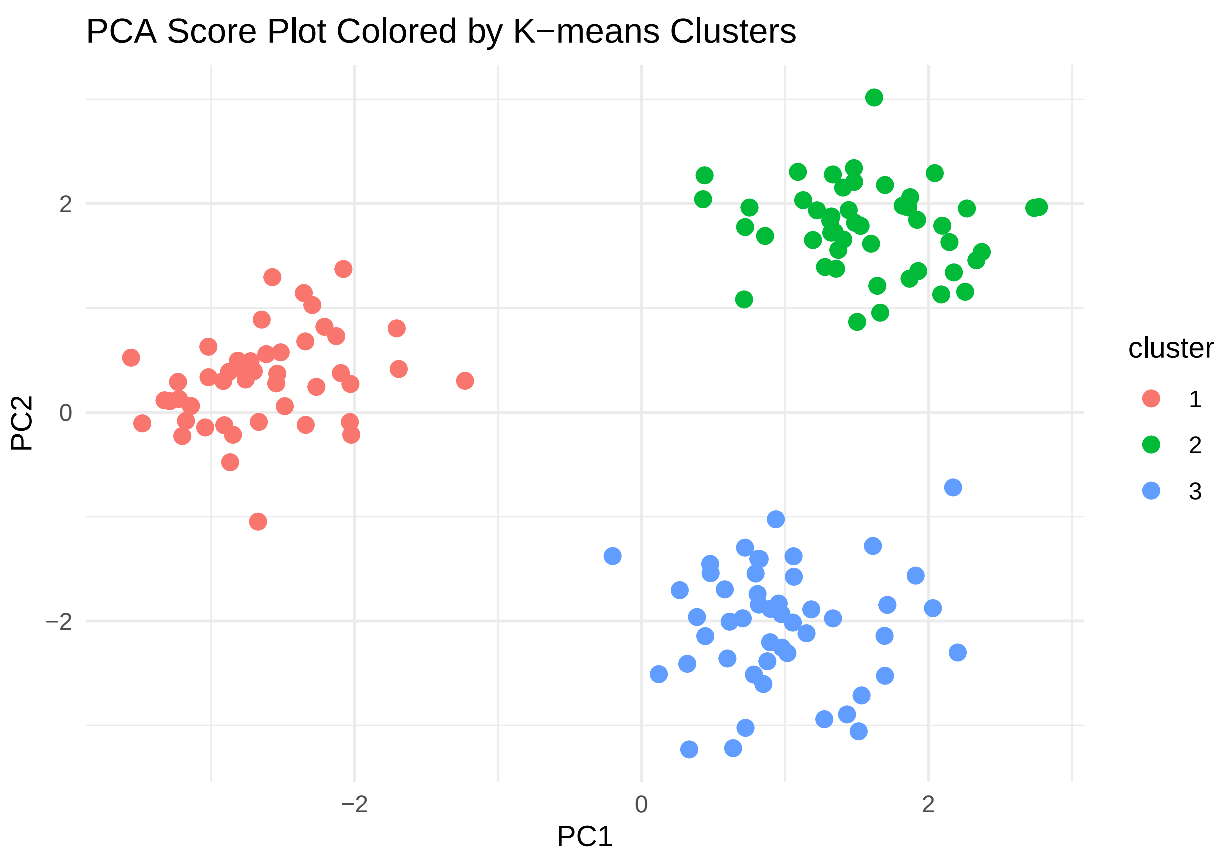
```r
pc2_abs
```

```
##      screen_time   exercise_days attendance_rate    social_hours     study_hours
##       0.55065059      0.49746524      0.44612633      0.32398583      0.27134228
##     stress_score     sleep_hours     caffeine_mg
##       0.19997246      0.16060015      0.07686821
```

```r
# Interpretation should refer to the variables with the strongest loadings.
# PC1 has positive loadings on sleep_hours and social_hours, and strong negative loadings on stress_sco
# PC2 is strongly driven by screen_time (negative), exercise_days (positive), and attendance_rate (posi

# Q10: PC1 vs PC2 colored by K-means clusters
pca_scores <- as.data.frame(pca_fit$x)
pca_scores$cluster <- factor(km_scaled$cluster)

ggplot(pca_scores, aes(x = PC1, y = PC2, color = cluster)) +
  geom_point(size = 2.5) +
  theme_minimal() +
  labs(title = "PCA Score Plot Colored by K-means Clusters")
```




```r
# PCA is a dimension reduction method that creates new variables (principal components)
# to summarize variation in the data, whereas clustering is a grouping method that
# assigns observations into clusters based on similarity.

# The PCA score plot shows strong separation among the three K-means clusters in the first two PCs, whi
```